2024 Third International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT) | 979-8-3503-6908-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICEEICT61591.2024.10718550

SliceNet: A Hybrid CNN Model for Network Slicing in 5G Communications

Soujanya S Badawadagi, Manoj Kumar. CM

Dept. of Electronics and Communication PES University, Bengaluru, India

soujanyabadawadagi@gmail.com, cmmanoj2001@gmail.com

***Abstract*—Network Slicing (NS) has emerged as a promising solution to address the challenges of reliability, ultra-low latency, and efficient resource allocation in 5G mo- bile networks. This paper introduces SliceNet, a novel model that employs a hybrid Convolutional Neural Network (CNN) to analyze network Key Performance Indicators (KPIs) and optimally allocate resources across network slices. By training on available KPIs, SliceNet learns to analyze incoming traffic and predict the most suitable network slice for unknown device types. The model's ability to make intelligent decisions ensures the selection of the optimal network slice, even in the event of ambiguous network conditions. SliceNet aims to enhance 5G network efficiency and reliability by dynamically allocating resources based on real-time conditions and device requirements, ultimately improving overall network performance and user experience.**

***Index Terms*—5G mobile networks, Wireless Communications, Network Slicing, Key Performance Indicators (KPIs), Convolutional Neural Network**

I. Introduction

In recent years, mobile communication has become a vital aspect of our daily lives, with the number of mobile devices growing at an unprecedented rate. This growth has been driven by the emergence of innovative services and applications that have transformed the way people communicate and interact with technology. However, managing these diverse technologies seamlessly has proven to be a significant challenge for heterogeneous wireless networks, despite the efforts of service providers to meet the ever-increasing demands of their customers.

The introduction of the next-generation 5G network, which builds upon the current LTE network, is set to revolutionize the wireless industry by opening up new business opportunities and driving innovation. 5G networks are designed to be multi-service platforms, capable of supporting a wide range of operations with varying performance requirements and services. This calls for a more extensive device ecosystem that can deliver enhanced mobile experiences, such as media mobilization, high-speed mobility, augmented reality, and connected vehicles, even in congested network environments.

To address these challenges, an approach that incorporates advanced techniques to analyze traffic patterns and requirements is proposed, enabling precise decision-making in 5G networks. By employing a hybrid Convolutional Neural Network (CNN) approach, the aim is to optimize network slice selection for devices and services, accurately predicting traffic demands and efficiently allocating resources.

Moreover, the emergence of technologies such as Software Defined Networks (SDN) and Network Function Virtualization (NFV) has further transformed the telecom industry, enabling new business models and enhancing customer experiences. Critical services expected to be encapsulated within 5G net- works include autonomous driving, AR-VR solutions, indus- trial automation, remote monitoring, smart health, and smart cities. Network slicing, considered a key enabling technology by the Third Generation Partnership Project (3GPP), allows operators to efficiently manage multiple network instances over a single infrastructure, catering to various applications, use cases, and business services while ensuring superior Qual- ity of Service (QoS).

The proposed model, SliceNet, focuses on two primary objectives: (1)Appropriate selection of a network slice for a device [1], and (2)Adaptive resource allocation in ambiguous network conditions. By leveraging a hybrid CNN approach,the aim is to achieve the most efficient and optimized selection of network slices, ultimately enhancing the overall performance and reliability of 5G networks.

In the following sections, the details of this work are elaborated. Section II provides a brief introduction to 5G network slicing, while Section III describes the dataset used in the study. Section IV explains the SliceNet model in detail, and Section V presents the results, discussing its application for slice prediction in scenarios involving unknown device types and network failovers.

II. 5G Network Slicing

5G network slicing [1] [2] is a fundamental architectural concept that enables the partitioning of a single physical network infrastructure into multiple virtual networks, each tailored to specific use cases, applications, or customer require- ments [3]. This segmentation enables the creation of dedicated network slices with tailored characteristics, including distinct Quality of Service (QoS) levels, latency parameters, and band- width allocations. In contrast, 4G LTE lacks this granularity, offering a one-size-fits-all approach to network management.

The customization offered by 5G network slicing addresses the shortcomings of 4G LTE, particularly in terms of flexibility and efficiency. With network slicing, operators can dynamically allocate resources based on changing demands, optimizing resource utilization and enhancing overall network performance. This dynamic allocation ensures that each network slice operates independently, without impacting the performance of other slices, thereby improving security, reliability, and privacy.

The implementation of network slicing in 5G networks relies heavily on two technologies: Software Defined Networking (SDN) and Network Function Virtualization (NFV). SDN provides centralized control and programmability of network resources [4], allowing operators to dynamically manage and allocate these resources as needed. On the other hand, NFV virtualizes network functions, enabling them to operate as software instances on standard hardware platforms. By combining SDN and NFV, operators can create, modify, and manage network slices in real-time without the need for extensive physical infrastructure modifications. Furthermore, SDN and NFV enable end-to-end orchestration of network slices, ensuring smooth coordination across various network domains and technologies.

III. DATASET

The dataset contains detailed information on various service types and network performance metrics in 5G communication [5]. It includes indicators for applications like IoT, AR/VR, Healthcare, and Public Safety, along with metrics such as packet loss rate and delay. This allows for analyzing the impact of different services on network performance, crucial for designing optimized 5G network slices.

eMBB (enhanced Mobile Broadband): Slice Type 1 (eMBB) targets high data rate applications needing significant capacity and coverage, ideal for urban areas and hotspots. Use cases include high-definition video streaming, AR/VR, and high-speed internet. This slice ensures high throughput, suited for data-intensive applications.

URLLC (Ultra-Reliable Low-Latency Communications): Slice Type 2 (URLLC) supports ultra-low latency and high reliability, crucial for mission-critical services like autonomous smart transportation, healthcare, and industry 4.0. It ensures minimal delay and maximum reliability for real-time communication needs.

mMTC (massive Machine Type Communications): Slice Type 3 (mMTC) is optimized for massive IoT connectivity, supporting numerous low-power devices. It is tailored for IoT applications such as smart cities, agriculture, and industrial IoT sensors, focusing on scalability and energy efficiency.

LTE/5G Categories: The LTE/5G Category column classifies usage scenarios by priority and performance needs. Categories **1-5** are for low-priority applications like web browsing. Categories **6-10** cover moderate-priority services like video streaming. Categories **11-15** are for high-priority applications needing low latency, such as AR/VR gaming. Categories **16-20** are reserved for mission-critical services like healthcare and public safety, ensuring these essential applications receive the necessary network resources and performance.

Train dataset: 31,583 entries and 17 columns
Test dataset: 31,585 entries and 16 columns

TABLE I. SLICE TYPES AND ASSOCIATED SERVICE TYPES

Slice Type	Associated Service Type	Characteristics	Applications
1	eMBB	High throughput, high data rates, improved capacity	AR/VR/Gaming, Smart Phones
2	URLLC	Low latency, high reliability	Smart city, IoT devices
3	mMTC	Massive connectivity, low power, scalable	Public Safety, Industry 4.0, HealthCare

IV. SLICENET MODEL

The SliceNet model is designed to leverage both spatial and temporal features from network traffic data to make accurate predictions about network slices. The input dataset [5] for the model includes various features such as LTE/5G category, time, packet loss rate, packet delay, and different application types like IoT devices, healthcare, smart city, and more. Each feature captures a crucial aspect of network performance and service requirements, allowing the model to understand the diverse demands on the network. The model architecture combines Convolutional Neural Networks (CNN) with Long Short-Term Memory (LSTM) networks.

The SliceNet architecture is as follows:

TABLE II. SLICENET ARCHITECTURE

Layer	Description
Input	Shape: (X train.shape[1], 1)
Conv1D MaxPooling1D	3x1, Stride=(1), ReLU, Padding=same, Filters: 64 Pool size=2
Conv1D MaxPooling1D	3x1, Stride=(1), ReLU, Padding=same, Filters: 128 Pool size=2
Conv1D MaxPooling1D	3x1, Stride=(1), ReLU, Padding=same, Filters: 256 Pool size=2
LSTM LSTM	Units: 128, Return sequences=True Units: 128
Dense Dropout	ReLU, Units: 128 Rate: 0.5
Dense	Softmax, Units: Num of Slice Types

To predict the final slice type using the given data, both spatial and temporal dependencies are essential. Spatial dependencies refer to the relationships among various features

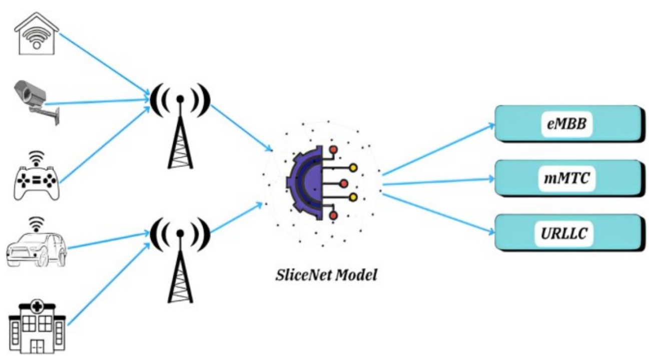

Fig. 1. SliceNet model overview

at a single point in time, such as the interaction between `PacketLossRate` and `Packetdelay`, which can indicate network quality issues, and categorical features like `LTE5gCategory`, `GBR`, and application-specific indicators `ARVRGaming`, `Healthcare`, which help determine the appropriate network slice. Temporal dependencies capture how these features evolve over time, allowing the model to identify trends and sequential patterns, such as changes in `PacketLossRate` or `Packetdelay`, which inform about the network's stability and performance. By leveraging Conv1D layers, the SliceNet model learns spatial dependencies, while LSTM layers capture temporal dependencies, enabling the model to accurately predict the slice type needed to meet network and service requirements dynamically. This integration ensures optimal performance and resource utilization in the network.

V. Experimentation and Results

In this section, the experimentation details are presented, along with a discussion of the results obtained from evaluating the SliceNet model for network slicing in 5G communications. Two experiments were conducted to assess the model's performance under different scenarios.

A. *Slice Prediction for Unknown Device Types*

In the first experiment, the goal is to evaluate the SliceNet model's ability to predict the appropriate network slice for unknown device types based on network performance metrics and service characteristics.

1) **Methodology:**

- **Dataset Preparation:** A comprehensive dataset containing detailed information on various service types, network performance metrics, and slice types in 5G communication is utilized.
  A key focus is to optimize memory usage to enhance computational efficiency as there exists limited resources in real time. This involved carefully assessing each column in the dataset to minimize memory usage while maintaining data integrity. Numerical data types are evaluated to determine if smaller representations, such as int32 or float32, could be employed without sacrificing precision. By selecting the most space-efficient data types for each column, memory overhead is significantly reduced.
  This optimization strategy enabled more effective utilization of computational resources, resulting in faster processing and smoother model execution.
- **Model Training:** The SliceNet model was trained on a subset of the dataset, consisting of 31,583 entries and 17 columns, including features such as LTE/5G category, time, packet loss rate, packet delay, and application types.
- **Evaluation:** The trained model was evaluated on a separate test dataset containing 31,585 entries to assess its performance in predicting network slices for unknown device types.

2) **Results**

- **Accuracy**: The SliceNet model achieved an accuracy of **96.97%** in predicting the correct network slice for unknown device types which is better than the Model proposed in Deep slice paper [1] .

- **Precision, Recall and F1 score** :The model demonstrated high precision, recall, and F1 score across different network slice types, indicating robust performance in slice prediction which is in comparison the existing models proposed [1] [2] .

The following table indicates the Classification report of Experiment 1 :

TABLE III. CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score
1	0.97	0.98	0.97
2	0.96	0.95	0.96
3	0.98	0.98	0.98

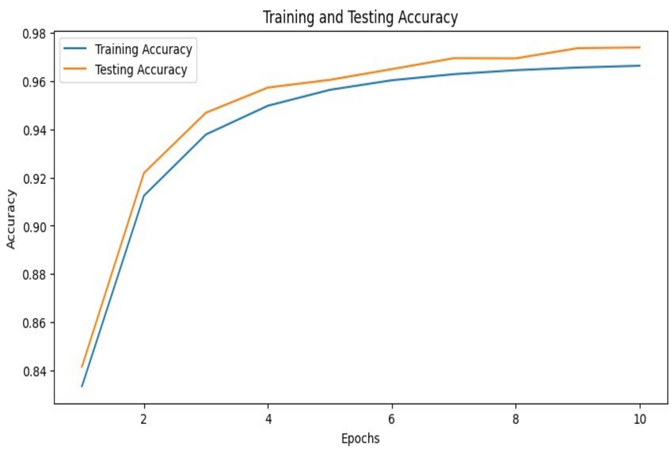

Fig. 2. Train and Test Accuracy

Building on these results, Experiment 2 will further enhance the model's capability by addressing ambiguous network conditions. We will introduce a 'master slice' to handle confusing entries more effectively, aiming to improve

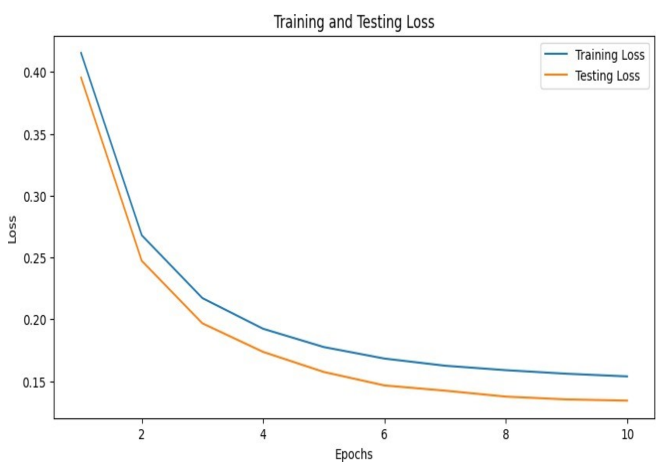

Fig. 3. Train and Test Loss

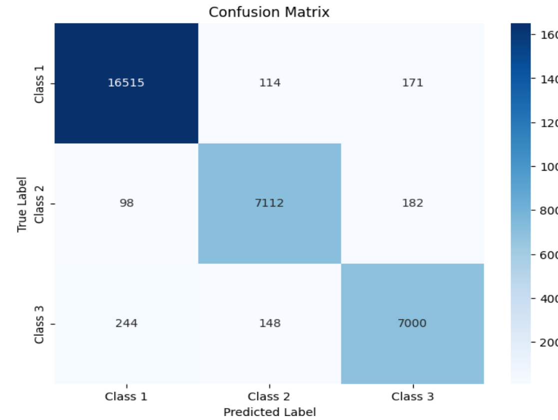

Fig. 4. Confusion Matrix for Experiment 1

overall classification performance and reduce misclassifications.

B. *Adaptive Resource Allocation in Ambiguous Network Conditions*

In the second experiment, the aim is to assess the SliceNet model's ability to adapt slice assignments during ambiguous network conditions, ensuring continuous service delivery and optimizing resource utilization.

For this experiment, the dataset is augmented with an additional 1000 samples in both the training and testing sets. These samples included variations in Key performance indicators and applications. Specifically, the applications were randomly generated as 1 or 0 for a slice type to challenge the model and assess its adaptability in scenarios where confusion might arise.

To enhance the model's adaptability, we introduced a strategic approach. All ambiguous entries were directed into a designated fourth slice called the "master slice." Within this master slice, the model focused solely on key parameters such as LTE/5G category, packet loss rate, and packet delay. Based on these parameters, the model intelligently assigned the entries to one of the three primary slices: eMBB, URLLC, or mMTC.

This approach is superior to direct slice assignment in scenarios where ambiguity exists due to the potential for misleading applications. Applications can often provide incomplete or misleading information, leading to confusion in slice assignment. For instance, an application may exhibit characteristics that overlap between different slice types or may not accurately reflect the underlying network requirements.

By directing ambiguous entries to a designated "master slice" and subsequently analyzing key parameters the model can make more informed decisions. This strategy allows the model to filter out irrelevant or misleading information from the applications, focusing instead on fundamental network metrics that better represent the underlying network conditions and service requirements.

- **Results :** The model achieved a significant improvement in accuracy, achieving a remarkable **97.6%**. The Precision, Recall, and F1 scores have remained consistent, underscoring the model's resilience in maintaining accurate predictions while navigating complex network environments

By implementing this method, the entries that were misleading in the previous experiment are effectively addressed, allowing the model to reevaluate and assign them to the appropriate categories reducing the overall errors.

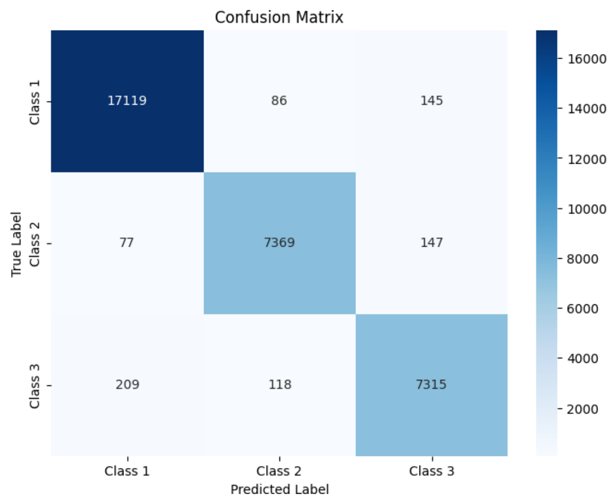

Fig. 5. Confusion Matrix for Experiment 2

VI. Conclusion

Network slicing in 5G is essential for next-generation wireless networks, offering significant benefits to mobile operators and businesses. This paper demonstrated the effectiveness of the SliceNet model in accurately predicting the optimal network slice based on key device parameters. Additionally, the model demonstrated adaptability to ambiguous network conditions using neural network techniques. Future efforts will focus on integrating SliceNet with emerging 5G technologies, exploring its potential in dynamic environments, and enhancing its capabilities to support advanced features such as real- time data analytics, predictive maintenance, and autonomous network management.

REFERENCES

[1] A. Thantharate, R. Paropkari, V. Walunj, and C. Beard, "DeepSlice: A Deep Learning Approach towards an Efficient and Reliable Network Slic- ing in 5G Networks," in 2019 IEEE 10th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York City, NY, USA, 2019.

[2] A. Thantharate, R. Paropkari, V. Walunj, C. Beard, and P. Kankariya, "Secure5G: A Deep Learning Framework Towards a Secure Network Slicing in 5G and Beyond," in 2020 10th Annual Computing and Communication Workshop and Conference (CCWC), Las Vegas, NV, USA, 2020.

[3] M. Shurman, E. Taqieddin, O. Oudat, R. Al-Qurran, and A. A. A. Nounou, "Performance enhancement in 5g cellular networks using priorities in network slicing," in 2019 IEEE Jordan International Joint Conference on Electrical Engineering and Information Technology (JEEIT), 2019, pp. 822–826.

[4] V. P. Kafle, Y. Fukushima, P. Martinez-Julia, and T. Miyazawa, "Consid- eration On Automation of 5G Network Slicing with Machine Learning," in 2018 ITU Kaleidoscope: Machine Learning for a 5G Future (ITU K), Santa Fe, Argentina, 2018.

[5] (2022) 5g slice type dataset. [Online]. Available: https://shorturl.at/7gqw